Modeling Bimanual Perceptual Interactions in Macaque Monkeys

Bayesian Hierarchical Modeling

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Outline

Task and Data

Model A: Sessions Pooled
    Model Structure

Model B: Sessions Hierarchical

Model C: Distractor Amplitude as a Covariate

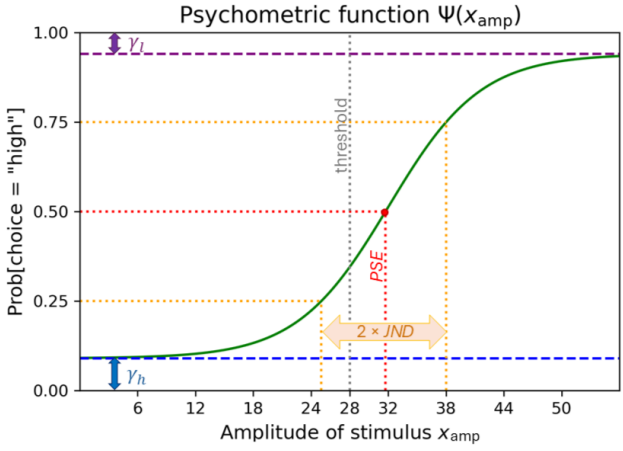
Summary

The task: 2AFC categorization of vibration amplitude

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\mu m$
  - $x_{\text{d}} \in \{6, 12, 18, 24, 32\}\mu m$ value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 μm.
  - "low" response: correct for $x_{\text{stim}} < 28\mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\mu m$.
- correct responses are rewarded with water or juice

Psychometric function: response $r \sim \text{Bernoulli}(p = \psi)$

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}})$$

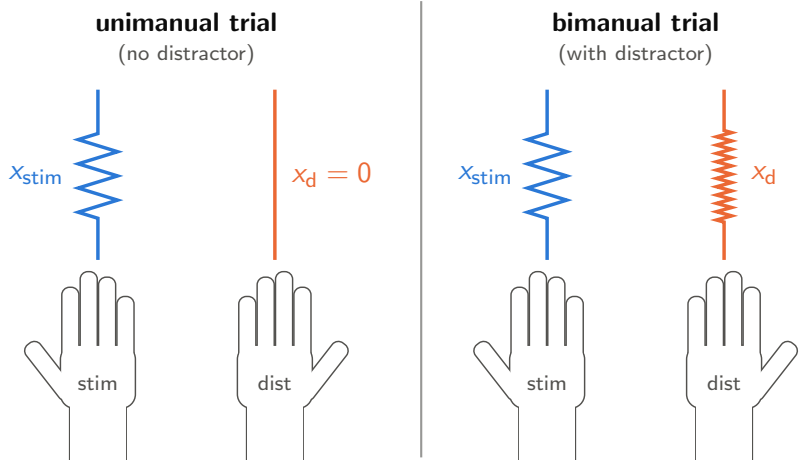


Parameters

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}}), \qquad \vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$$

- γ_h: lapse rate for "high"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"high"})$, even to the weakest stimuli
  - lower (left) asymptote of ψ
- γ_l: lapse rate for "low"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"low"})$, even to the strongest stimuli
  - $1 - \gamma_l$ = upper (right) asymptote of ψ
- logit^{-1}: the logistic sigmoid, $\text{logit}^{-1}(z) = \dfrac{1}{1 + \exp(-z)}$
- PSE: stimulus where responses of "high" and "low" are equally likely
  - x_{stim} such that $\psi(x_{\text{stim}}) = 0.5$
  - represents bias
- JND: how much the stimulus needs to change to be noticed half the time
  - half the distance between $\psi(x_{\text{stim}}) = 0.25$ and 0.75
  - represents sensitivity

The task: one trial



Modeling bimanual interactions

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  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\mu m$
  - $x_{\text{d}} \in \{6, 12, 18, 24, 32\}\mu m$ value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 μm.
  - "low" response: correct for $x_{\text{stim}} < 28\mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\mu m$.
- correct responses are rewarded with water or juice

The data

- One monkey (Sirius), 12,937 trials.
- Collected over 42 sessions, averaging approximately 308 trials per session.
- 8 stimulus amplitudes, 6–50 μm; 4 groups: hand $\times$ (unimanual, bimanual).
- Each session uses a single distractor amplitude (6–32 μm) on its bimanual trials.

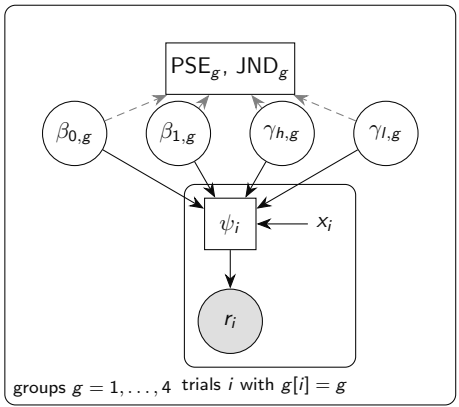
	left hand		right hand		
x_{stim} (μm)	unimanual	bimanual	unimanual	bimanual	total
6	408	404	404	402	1,618
12	407	400	408	400	1,615
18	402	400	403	405	1,610
24	402	405	402	406	1,615
32	404	405	405	407	1,621
38	403	408	404	408	1,623
44	406	403	404	405	1,618
50	405	405	406	401	1,617
total	3,237	3,230	3,236	3,234	12,937

Three models

Model	Sessions	Distractor
A	pooled: one curve per group	uni and bi fit separately
B	hierarchical: one curve per group and session	uni and bi fit separately
C	hierarchical, one group per hand	covariate: shift $\beta_2 d$

For each model: convergence study on synthetic data, then the fit to the real data.

Model A: graphical model



- circles: random variables
- shaded: observed
- rectangles: deterministic
- x_i: fixed covariate
- dashed: derived quantities, not part of the likelihood

Model A: likelihood

For trial i in group $g[i] \in \{\text{left_uni},\ \text{left_bi},\ \text{right_uni},\ \text{right_bi}\}$:

$$r_i \sim \text{Bernoulli}(\psi_i), \qquad \psi_i = \gamma_{h,g[i]} + (1 - \gamma_{h,g[i]} - \gamma_{l,g[i]})\ \text{logit}^{-1}(\beta_{0,g[i]} + \beta_{1,g[i]}\, x_i)$$

- $r_i = 1$ for "high", 0 for "low"
- x_i: stimulus amplitude, standardized by the mean and sd of the 8 levels
- One psychometric curve per group, shared by all 42 sessions: $4 \times 4 = 16$ parameters
- ψ_i clipped to $[10^{-8},\ 1 - 10^{-8}]$ for numerical stability

TODO: one-line motivation: simplest model, a baseline

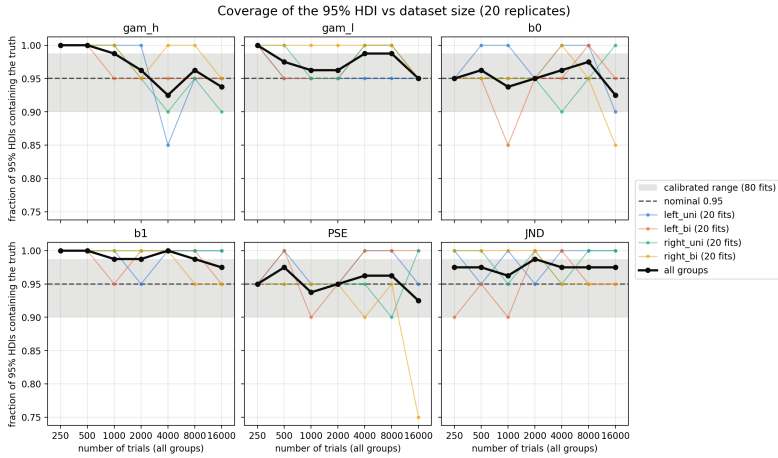
Model A: priors

The same independent priors in every group g:

$$\begin{aligned}
\beta_{0,g} &\sim \mathcal{N}(0,\ 6) \\
\beta_{1,g} &\sim \mathcal{N}(4,\ 2) \\
\gamma_{h,g} &= 0.25\, w_{h,g}, \qquad w_{h,g} \sim \text{Beta}(1.22,\ 4.89) \\
\gamma_{l,g} &= 0.25\, w_{l,g}, \qquad w_{l,g} \sim \text{Beta}(1.22,\ 4.89)
\end{aligned}$$

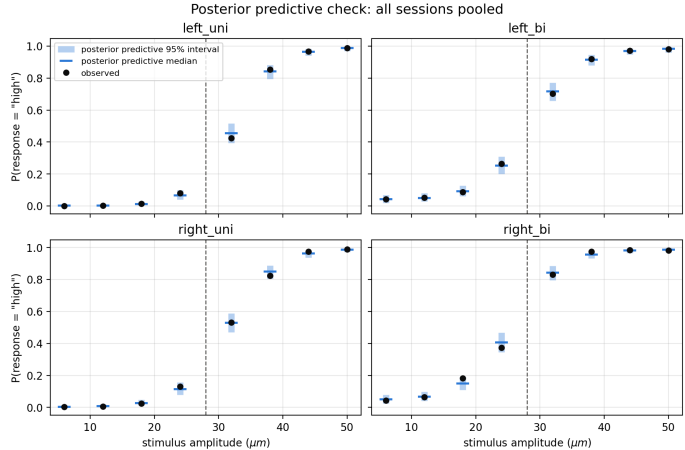
- Beta with mean 0.2, sd 0.15; scaling by 0.25 keeps each lapse rate in $(0, 0.25)$, so ψ is increasing and the PSE and JND exist
- Implied lapse rate: mean 0.05, 95% range [0.003, 0.141]
- Not checked by prior predictive simulation

Model A: convergence on synthetic data



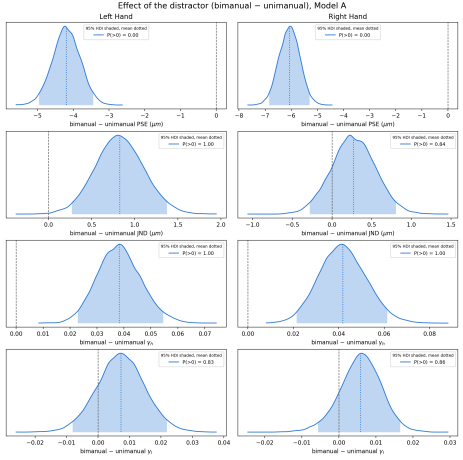
TODO: takeaway; numbers from Convergence_A_Report, "Conclusions"

Model A: fit to the real data



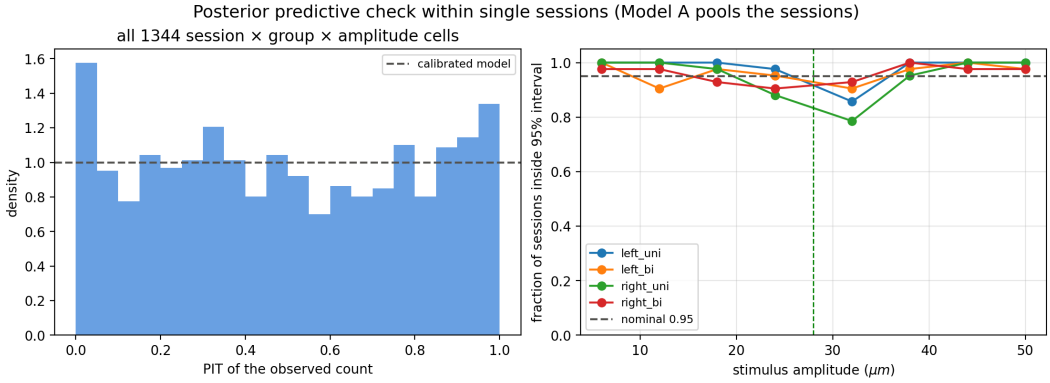
TODO: pooled check passes: all 32 proportions inside (Fit_A_Report, "Model Checks")

Model A: effect of the distractor



TODO: PSE shift, γ_h and JND effects (Fit_A_Report, Summary)

Model A misses the variation between sessions



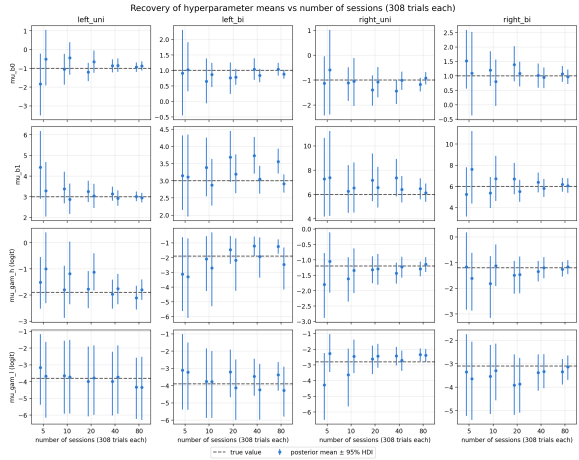
TODO: within single sessions responses vary more than A predicts (Fit_A_Report, "Model Checks")

Model B

- A curve for every group *and session*, drawn from group-level distributions (means and between-session sds).
- Design-based priors, checked by prior predictive simulation.

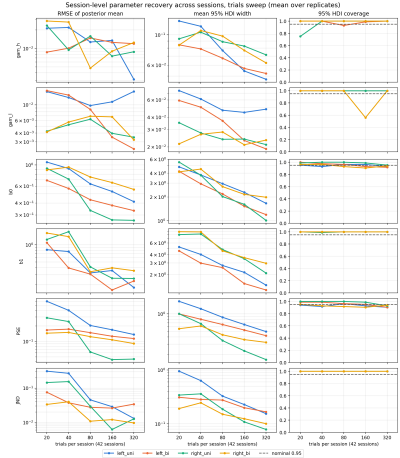
TODO: plate diagram or equations (Convergence_B_Report, "Model Structure")

Model B: convergence on synthetic data



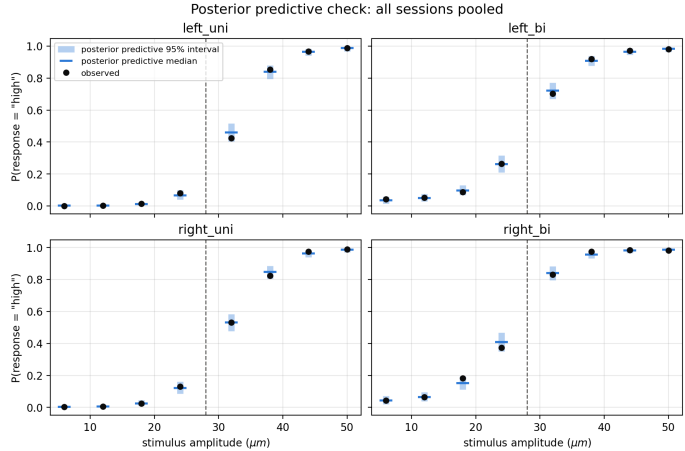
TODO: what is recovered at the real data size (Convergence_B_Report, "Conclusions")

Model B: per-session recovery



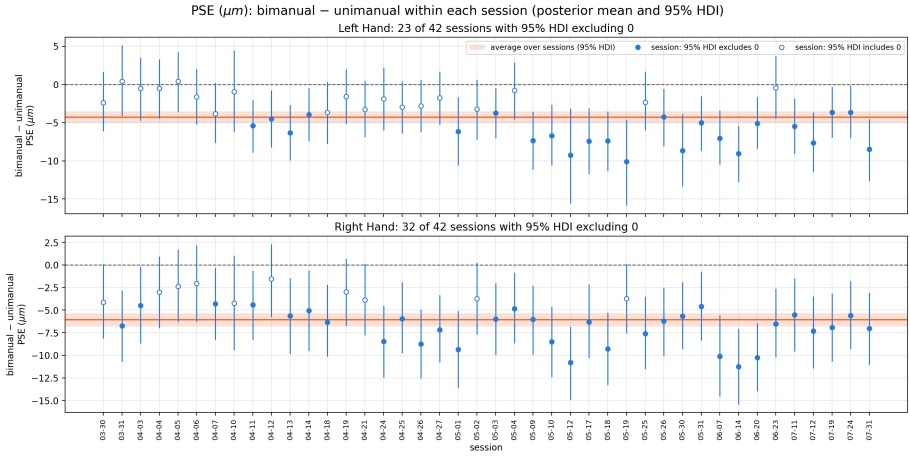
TODO: per-session PSE and JND reliable at ∼300 trials

Model B: fit to the real data



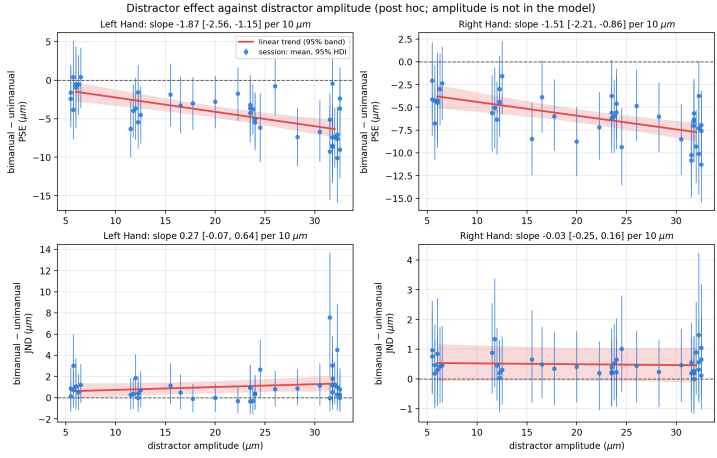
TODO: all 32 proportions inside; 99.8% of session cells (Fit_B_Report, "Posterior predictive checks")

Model B: PSE shift in every session



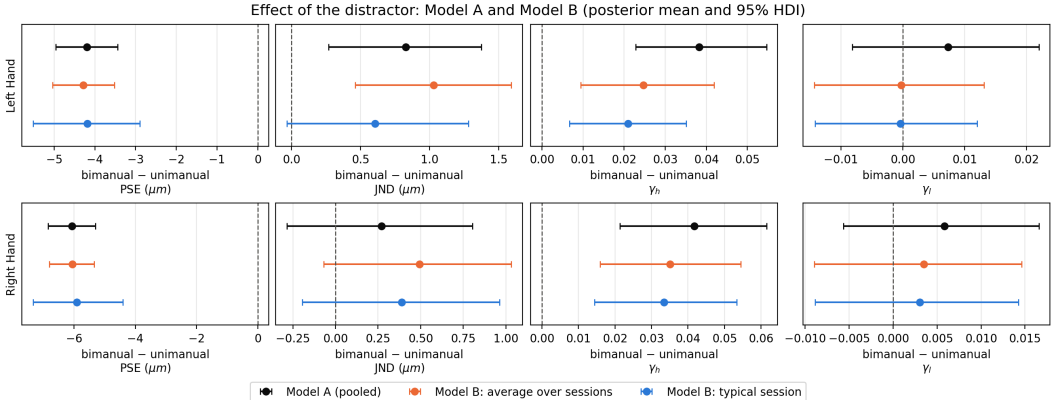
TODO: shift negative in 40/42 (left) and 42/42 (right) sessions (Fit_B_Report, Summary)

Model B: the shift grows with distractor amplitude (exploratory)



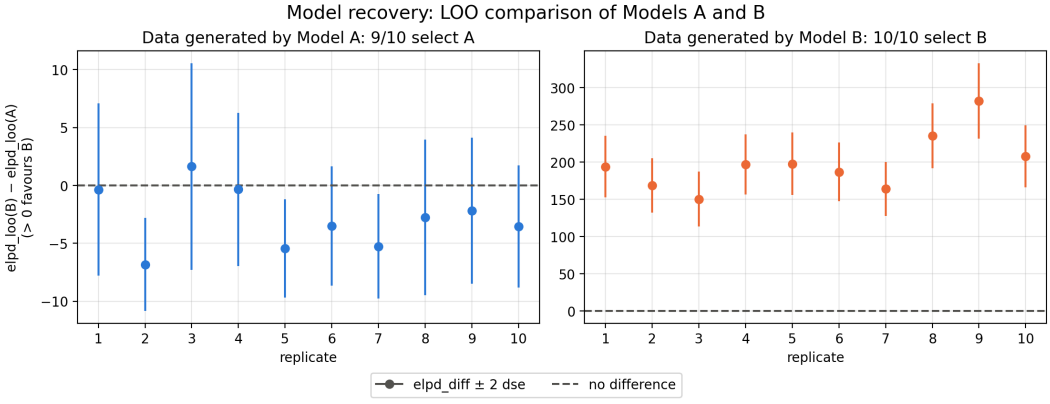
TODO: about -2 μm per 10 μm; confounded with session order

Model B against Model A



TODO: same effects; B better by LOO, 164 (SE 19) (Fit_B_Report, "Comparison with Model A")

Is B's advantage real? Model recovery



TODO: B wins on all data from B, ties or loses on data from A (Model_Recov_BA_Report)

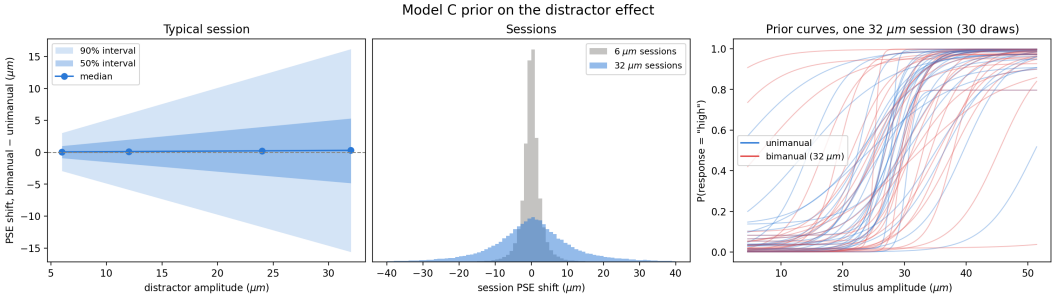
Model C

$$\psi(x, d) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x + \beta_2 d)$$

- d: distractor amplitude (0 on unimanual trials); one group per hand.
- Uni- and bimanual trials share slope and lapse rates: the distractor only shifts the curve, by an amount proportional to d.

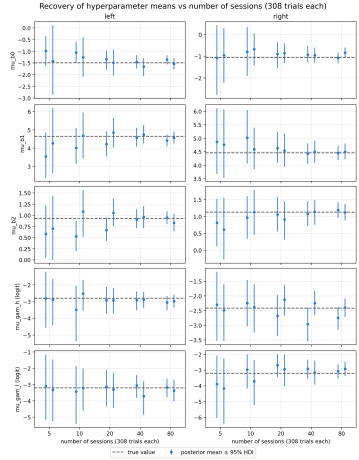
TODO: why: test the amplitude trend from Model B inside a model

Model C: prior on the distractor effect



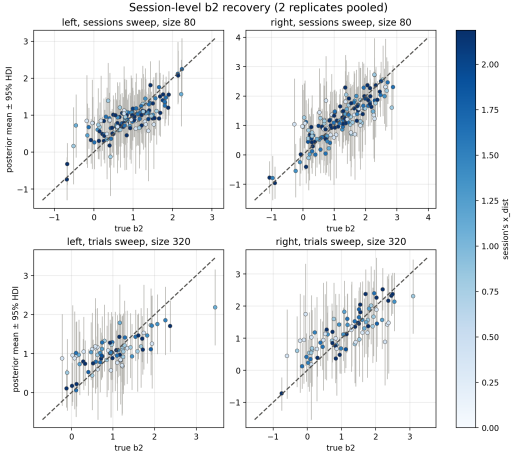
TODO: weakly informative; a weak distractor can only shift a little (Convergence_C_Report)

Model C: convergence on synthetic data



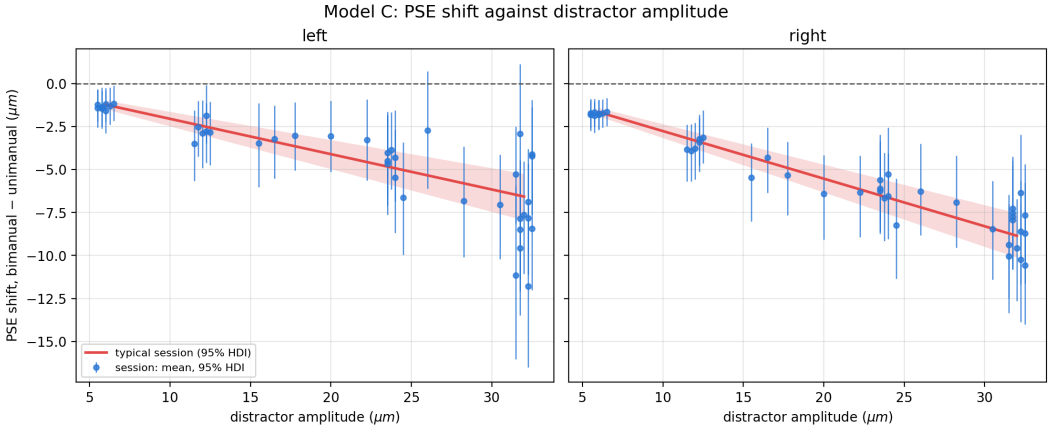
TODO: all hand-level means recovered, μ_{β_2} well (Convergence_C_Report, "Conclusions")

Model C: per-session β_2 is weak for small distractors



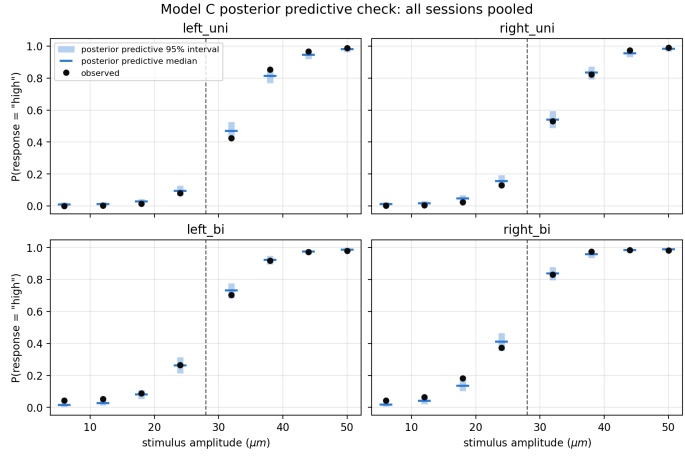
TODO: shrinkage toward the hand mean at 6 μm

Model C: the shift grows with distractor amplitude



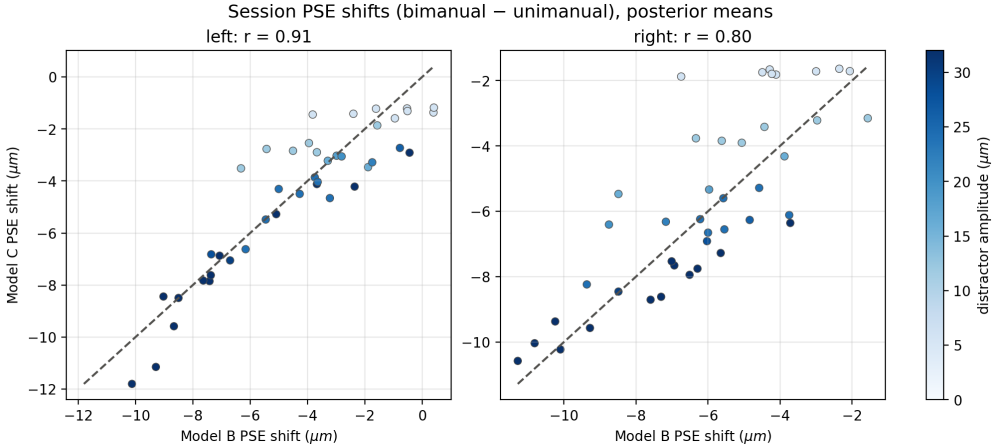
TODO: μ_{β_2} and the typical shift at 6 and 32 μm (Fit_C_Report, "Effect of the Distractor")

Model C misfits at the weakest stimuli



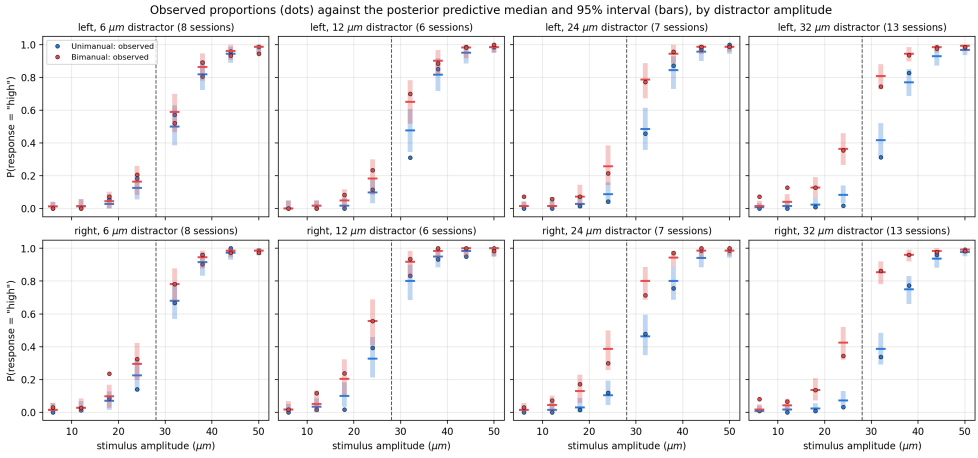
TODO: shared lapse rates: 6 of 32 proportions outside (Fit_C_Report, "Posterior predictive check")

Model C against Model B



TODO: LOO favors B by 37 (SE 12); 6 μm shifts too small (Fit_C_Report, Comparison)

Model C: shift by distractor amplitude, observed vs predicted



TODO: right hand at 6 μm outside: a presence effect (Fit_C_Report, amplitude check)

Comparison of the models

	A	B	C
Sessions	pooled	hierarchical	hierarchical
Distractor	separate groups	separate groups	covariate $\beta_2 d$
elpd_{loo}	**TODO:**	**TODO:**	**TODO:**
PSE shift, left / right (μm)	**TODO:**	**TODO:**	**TODO:**

TODO: fill from Fit_A/B/C_Report; elpd differences from Fit_B and Fit_C

Conclusions

- The distractor lowers the PSE: **TODO:** size, both hands.
- It raises the rate of "high" responses to weak stimuli (γ_h).
- The sessions differ; a hierarchical model is needed.
- The shift grows with distractor amplitude, **TODO:** but not proportionally?

Next steps

- Revise Model C: lapse rates per condition, a distractor-presence term, possibly a bimanual slope.
- Model recovery for B against the revised C.
- **TODO:** further data or experiments